NFS uses the nfs-secure service to help negotiate and manage communication with the server when connecting to Kerberos-secured shares. It must be running to use the second NFS shares; start and enable it to ensure it is always available.

```
# systemctl enable nfs-secure
# systemctl start nfs-secure
```

Note: The *nfs-secure* command is part of the *nfs-utils* package, which should be installed, by default. If it is not installed, use the following command:

```
# yum -y install nfs-utils
```

Exporting NFS

The Network File System (NFS) is commonly used by UNIX systems and network attached storage devices to allow multiple clients to share access to files over the network. It provides access to shared directories or files from client systems.

NFS exports

An NFS server installation requires the *nfs-utils* package to be installed. It provides all the necessary utilities to export a directory with NFS to clients. The configuration file for the NFS server exports the */etc/exports* file. This file lists the directory to share to client hosts over the network, and indicates which hosts or networks have access to the export.

Note: Instead of adding the information required for exporting directories to the */etc/exports* file, a newly created file named **.exports* can be added to the */etc/exports.d/* directory holding the configuration of exports.

Warning: Exporting the same directory with NFS and Samba is not supported on Red Hat Enterprise Linux 7, because NFS and Samba use different file locking mechanisms, which can cause file corruption.

One or more clients can be listed, separated by a space, as any of the following:

1. DNS-resolvable host name, like *server.example.com* in the following example, where the */myshare* directory is exported and can be mounted by *serverX.example.com.*

```
/myshare        server.example.com
```

2. DNS resolvable host name with the wildcards '*' for multiple character and '/' or '?' for a single character. The following example allows all sub-domains in the *example.com* domain to access the NFS export:

```
/myshare     *.example.com
```

3. DNS resolvable host name with character class lists in square brackets. In this example, the hosts *server1.example.com, server2.example.com,…*and *server20.example.com* have access to the NFS export.

```
/myshare    server[0-20].example.com
```

4. IPv4 address. The following example allows access to the */myshare* NFS share from the *172.25.21.21* IP address.

```
/myshare    172.25.21.21
```

5. IPv4 network. This example shows a */etc/exports* entry, which allows access to the NFS-exported directory */myshare* from the *172.25.0.0/16* network.

```
/myshare    172.25.0.0/16
```

6. IPv6 address without square brackets. The following example allows the client with the IPv6 address *2000:472:18:b51:c32:a21* access to the NFS exported directory */myshare.*

```
/myshare    2000:472:18:b51:c32:a21
```

7. IPv6 network without square brackets. This example allows the IPv6 network *2000:472:18:b51::/64* access to the NFS export.

```
/myshare        2000:472:18:b51::/64
```

8. A directory can be exported to multiple hosts simultaneously by specifying multiple targets with their options, separated by spaces following the directory to export.

```
/myshare   *.example.com 172.25.0.0/16
```

Optionally, there can be one or more export options specified in round brackets as a comma-separated list, directly followed by each client definition. Three commonly used export options are given below.

ro, read-only: This is the default setting when nothing is specified. It is allowed to explicitly specify it with an entry. This restricts the NFS clients to read files on the NFS share. Any write operation is prohibited. The following example explicitly states the *ro* flag for the *server.example.com* host.

```
/myshare    desktop.example.com(ro)
```

rw, read-write: This allows read and write access for the NFS clients. In the following example, the *desktop.example.com* is able to access the NFS export read-only, while *server[0-20].example.com* has read-write access to the NFS share.

```
/myshare desktop.example.com(ro) server[0-20].example.com(rw)
```